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SECTION 1

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Development of an Apache Spark-Based Data Pipeline for Bike Sharing Demand Prediction Using Multi-Source Data Integration

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Abstract—Bike-sharing systems generate large volumes of data from daily user activities. The operators can perform data-driven demand forecasting and operational improvement by utilizing this data. However, integrating a large number of data from daily bike usage and transforming it into meaningful analytical information still remains challenging. This project develops an Apache Spark-based data pipeline for bike sharing demand by ingesting and integrating multiple data sources, such as Divvy bike trip records, weather information and public holiday data. The proposed pipeline follows a Data Lakehouse architecture consisting of Bronze, Silver and Gold layers to support data ingestion, transformation, integration and feature engineering. After that, a Random Forest Regression model is implemented to predict the bike demand using different features. The model shows a strong predictive capability by achieving an R^2 score of approximately 0.901. Finally, the processed data is further structured into a star schema and connectED to Power BI to provide interactive visualizations for demand analysis and assist the operators in the decision making process.

Keywords—*Apache Spark; ETL; Data Pipeline; Bike Sharing; Demand Prediction*

I. INTRODUCTION

A. Business Domain

This project focuses on the transportation and urban mobility domain, specifically bike-sharing services. In big cities with heavy traffic, for instance, Chicago and Evanston, bike-sharing systems have become an important part of smart city transportation by providing an environment friendly and convenient mode of travel. Divvy, the bike and scooter-sharing system generates large volumes of travel data daily, which can be used to gain insight into user behaviour and mobility patterns.

B. Problem Statement

Bike-sharing system, like Divvy, generates large volumes of data from multiple sources by collecting historical trip data,

including trip duration, start and end times, station locations and user types. However, although the dataset is rich in volume, the external environmental factors attributes, such as weather and holiday that can influence demand patterns for bike trips are limited. As a result, analysis is primarily based on internal historical patterns, which may limit the ability to understand demand patterns driven by external factors.

Additionally, the dataset mostly uses timestamp-based data instead of temporal features such as day of the week and monthly trends. Raw timestamp data alone is unable to accurately represent underlying usage patterns without appropriate feature engineering and transformation processes. Consequently, important information contained in the data may not be fully utilized during forecasting and analysis.

Therefore, there is a need for a structured data processing and feature engineering pipeline that transforms raw trip records into analytical datasets. By incorporating contextual factors, more meaningful analysis on bike usage patterns can be done to support data driven decision making.

C. Problem Relevance and Importance

The ability to analyze and predict bike-sharing demand accurately is important for improving sustainability in urban transportation systems. By understanding the usage patterns, bike-sharing operators can make informed decisions about bike allocation, service and station planning. Without accurate demand prediction, operators may experience problems like bike shortages at high-demand stations or underutilized resources at low-demand locations, which could cause user dissatisfaction and affect operational effectiveness.

As bike-sharing systems continue to generate large volumes of data every month, the need to transform raw bike record data into meaningful information is increasing to support operators' decision making process. However, the data is useless without proper processing, integration and feature

extraction techniques. Therefore, a structured data pipeline is crucial for extracting hidden patterns and trends from the data.

Furthermore, incorporating contextual factors that influence bike usage such as weather conditions and public holidays can provide a more comprehensive result of demand and trend prediction. The forecasting accuracy will be improving and changes in demand can be better anticipated. In conclusion, addressing this problem can enhance analytical capabilities while demonstrating the practical application of data engineering techniques in the bike-sharing domain.

D. Project Goal

The main goal of this project is to develop an Apache Spark-based data pipeline that integrates multiple data sources from Divvy Trip Data, Weather Data API and Holiday Dataset to support bike-sharing demand prediction.

E. Project Objectives

The objectives of this project are:

- 1) To integrate bike trip, weather and public holiday datasets into a unified data repository for analysis.
- 2) To develop an automated data pipeline using Apache Spark for data ingestion, processing and transformation.
- 3) To implement a machine learning model to predict bike-sharing demand based on historical data and external factors such as weather and public holiday.

II. LITERATURE REVIEW

A. Data Pipeline Concept

Data pipelines serve as the foundation for processing, analysis and decision making processes, especially in data-driven environments [1]. These pipelines facilitate the movement of data following a series of processes that extract, transform and load data from diverse sources to designated target systems. Through the well-organized data orchestration across different stages, an effective data pipeline framework can be established.

In addition to data movement, modern data pipelines play an important role in improving processing efficiency through automation, reducing the need of data being handled manually. Automated workflows, such as scheduled or batch processing tasks, can significantly reduce the time of data processing and human intervention to avoid data errors caused [2]. As a result, data quality and accuracy can be enhanced during the transformation stage in the data pipeline.

In an ideal design, the data pipeline should be flexible to adapt any data situation smoothly, especially for the data from multiple sources [3]. Figure 1 illustrates a simplified data pipeline architecture, consisting of three main stages: Extract, Transform and Load (ETL), incorporating downstream analytics and insight generation layers. This structure is commonly used in a data engineering system to build a data pipeline, aligning with modern data engineering practices to support data-driven decision making.

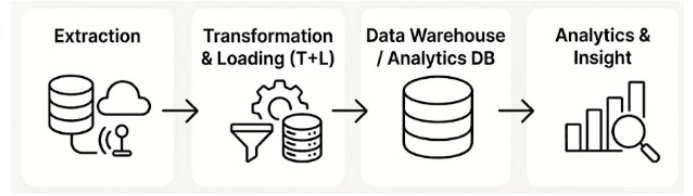


Figure 1. End-to-End Data Pipeline Architecture with Analytics Layer

B. Data Engineering in Bike-Sharing Domain

Bike-sharing systems are a form of shared mobility services to individuals, which is free of charge for a short term basis at a minimal rate and also allow users to rent bicycles for short distance travel. The majority of bike-sharing systems allow users to borrow and return bikes from any bike station that belongs to the same network. This domain has gotten a lot of attention in recent years due to its ability to minimize the negative impact on the environment and provide a new mode for transportation activities. As a result, it establishes a larger cycling community that minimizes greenhouse gas emission and indirectly reduces traffic issues and promotes healthier lifestyles among urban populations [4].

As bike-sharing services continue to expand, they generate a large volume of data on a daily basis, including trip records, station information, bicycle availability and user activity data. In addition to operational data, external factors such as weather conditions and holiday flags also influence the bike usage demand prediction. Previous studies have shown that environmental and temporal factors can significantly affect bike-sharing demand and should be considered when developing forecasting models [4]. However, these datasets are often collected from different sources and stored in different formats, which makes data integration and analysis challenging. Therefore, operators may face difficulties in managing the data without an efficient framework.

To address these challenges, data engineering plays an important role by enabling the automated collection, integration, processing and storage of data from multiple sources. Through automated data pipeline, raw data can be transformed into structured and analysis-ready datasets to support predictive analytics using machine learning algorithms. In addition, by implementing artificial intelligence within the data pipeline, tasks such as data quality checking and analysis can be done easier [5]. Modern data engineering technologies such as Apache Spark are particularly suitable for this purpose because it can handle large-scale historical data efficiently. Integration of the weather information and public holidays records also increase the accuracy of forecasting models. Organizations may create more precise demand forecast models and facilitate data-driven decision-making by combining bike trip records with external data sources like weather reports and public holiday calendars.

Therefore, developing a scalable data pipeline that integrates bike trip records with external datasets like weather and holiday information is essential to support accurate demand prediction and decision making in a bike-sharing system. By utilizing Apache Spark to automate data ingestion,

processing, storage and predictive analytics, the project perfectly addresses this need.

C. Feature Engineering for Bike-Sharing Demand Prediction

Feature engineering is important in transforming raw data into meaningful variables that can improve the usability of predictive models by improving data quality and providing relevant information to predictive models to generate accurate forecasting. In a bike-sharing system, raw operational records usually contain timestamp information and rental records which do not provide meaningful insight without further transformation before analysis steps. Sathishkumar et al. demonstrated that in order to capture variations in user demand patterns, temporal attributes in bike-sharing datasets can be converted into structured features, such as hour, weekday and seasonal indicators [4].

Besides, environmental factors such as weather condition, also considered important variables in demand prediction. Temperature, humidity, wind speed and rainfall can influence users' interest to rent bicycles and therefore affect overall demand patterns [4]. By integrating these external factors into raw bike-sharing historical datasets, the prediction model can better understand the relationship between environmental conditions and bicycle usage behaviour among all users.

D. Random Forest Regression for Demand Prediction

Machine learning algorithms are commonly used in bike-sharing systems to predict future demand based on historical usage patterns and influencing factors such as environmental factors mentioned in last section. Predictive modelling helps operators to estimate bicycle requirements in advance, to support better resource allocation and planning. Sathishkumar et al. applied data mining techniques to predict hourly bike-sharing demand using historical rental records with environment variables to understand complex bicycle usage demand and user behaviour.

Among various machine learning techniques, Random Forest is widely used for prediction problems involving multiple influencing factors because it can capture complex relationships between input variables and output values. Random Forest combines multiple decision trees to generate a final prediction model, which can handle non-linear relationships in real-world datasets. For bike-sharing demand prediction, factors like time, weather and seasonal patterns may influence and interact with each other. Therefore, Random Forest is suitable for modelling these variations.

Although Sathishkumar et al. focused on demand prediction rather than data pipeline architecture, the study highlights the importance of preparing high-quality structured datasets before applying any machine learning models [4]. Therefore, integrating Random Forest regression into a data pipeline enables processed data to be transformed into predictive insights that support decision-making in bike-sharing operations..

E. Data Visualization and Decision Support

After data processing and analysis, visualization is crucial to present extracted insights to support decision-making. Mobility datasets after processing usually contain complex

temporal and spatial patterns that are difficult to interpret without suitable visualization techniques. To overcome this challenge, interactive dashboards are used for the users to explore trends, compare different conditions and identify operational patterns.

De Godoi et al. discussed the use of Power BI dashboards to analyse processed traffic data after completing the ETL process. The visualization layer showed the identification of spatial and temporal patterns, such as areas with higher traffic occurrences and periods with increased activity levels [7]. This also proves that visualization can convert processed transportation data into understandable information efficiently for decision makers.

Similarly, in a bike-sharing system as in the project, dashboards can be used to display demand trends, rental distribution, peak usage period and the relationship between bicycle usage and external factors such as temporal and environmental factors. By integrating visualization into the data pipeline, operators can obtain actionable insights regarding bicycle allocation and system optimization.

III. METHODOLOGY

A. Methodology Selection

The system development approach used in this project is the CRISP-DM (Cross Industry Standard Process for Data Mining). It consists of six structured frameworks to guide data mining from business understanding to deployment [8], to ensure systematic flow in the development process.

CRISP-DM is selected because the proposed project involves both data engineering and machine learning components. This framework provided structured workflow that aligned well with the objectives of this project, that include data integration, data preparation, modeling and evaluating their performance.

The first phase, business understanding, focuses on accessing the business situation to get an overview of the project objectives and identify the problem of predicting bike demand. The second phase, data understanding, involves the process of collecting data from data sources, divvy bike trip data, weather data and holiday data, and exploring the data for data quality checking. The third phase, Data Preparation, includes data cleaning, integration and derived necessary attributes in the bike trip data to prepare for further analysis.

The fourth phase, Modeling, applies data mining techniques to predict the bike trip demand using historical and external factors data. The fifth phase, Evaluation, accesses the model performance using appropriate metrics to ensure accuracy of modeling. The final phase, Deployment, focuses on presenting the result in the final report and showing it through Power BI to support decision-making.

Figure 2 shows the CRISP-DM methodology employed in this project. The framework consists of six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation and Deployment, which guide the project from problem identification to the delivery of actionable insights.

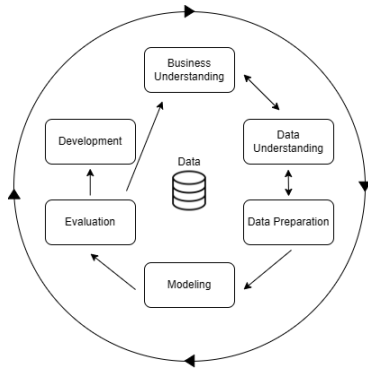


Figure 2. CRISP-DM Methodology

B. Business Understanding

Bike-sharing system generates a large amount of operational data from daily user activities. However, bicycle demand fluctuates depending on multiple factors, including historical usage patterns, weather conditions and temporal factors, such as weekdays, seasons and public holidays. Without proper analytical approach, operators may face difficulties in predicting future demand of bicycle usage and optimizing operational strategies to ensure bicycle availability.

Therefore, the project aims to develop a scalable data pipeline by utilizing Apache Spark to integrate historical bike-sharing records with external datasets (weather data and public holiday data) to produce an analytical dataset for demand prediction. A Random Forest Regression model is developed to forecast daily bike demand, while Power BI is used to visualize the prediction results.

C. Data Understanding

In this phase, datasets required for developing the bike-sharing demand prediction model are identified and extracted. This project consists of three main data sources:

1) Data Source 1: Divvy Trip Data (Open Dataset)

a) Type: Structured public dataset (CSV files)

b) *Description:* This dataset contains Cyclistic's bike-sharing trip records in Chicago, United States by both casual riders and annual members. Key attributes include `ride_id`, `rideable_type`, `start and end timestamp`, `start and end station information`, and `type of user membership`. This dataset serves as the primary source of user behaviour, trip duration patterns and trends.

c) *Relevance:* This is the core dataset for building the fact table (FactTrips) in the data warehouse. By using this dataset, rider behaviour differences between casual riders and members can be analyzed. Time and location based analytics are also supported using Spark ETL processing.

2) Data Source 2: Open-Meteo Weather Dataset

a) *Type:* External REST API (JSON-based weather data)

b) *Description:* The API provides historical weather data from 1940 and allows retrieval of hourly and daily weather variables, such as temperature, precipitation, wind speed and weather conditions. In this project, weather data

from Chicago that is aligned with bike trip timestamp is extracted to enrich the primary dataset with environmental factors.

c) *Relevance:* Weather conditions can influence user behaviour and bike-sharing demand. By integrating weather data with Cyclistic trip records, the project can further analyze how environmental conditions can affect user type behaviour and duration.

3) Data Source 3: United States Public Holiday Dataset

a) Type: Structured calendar dataset (CSV format)

b) *Description:* This dataset contains official US public holiday records, including date, name of holiday, weekday name and calendar breakdown. It can provide the national holiday information on a particular day that may influence user travel behaviour.

c) *Relevance:* Holiday periods often affect commuting patterns and bike usage demand. After integrating this dataset, peak usage on holidays and weekends can be identified and time-based analysis is supported.

C. Data Preparation

In this phase, raw datasets will be transformed into structured analytical data to perform further machine learning operations.

Apache Spark is used to perform data ingestion, transformation, integration and feature engineering because it is able to process large amounts of data efficiently. Apache Parquet is adopted as the storage format because its columnar structure enables efficient storage, faster query performance and supports both machine learning and business intelligence workloads.

The data preparation process follows a Data Lakehouse architecture consisting of Bronze, Silver and Gold layers. The Gold layer is further structured into a multidimensional database using a star schema to support business intelligence visualization.

1) Bronze Layer: Data Ingestion

a) This layer stores raw datasets collected from different sources without any modification. The original formats of the datasets are maintained.

b) The raw datasets include:

- 72 monthly Divvy trip CSV files
- Daily weather dataset
- Public Holiday Dataset

2) Silver Layer: Data Processing and Transformation

a) The Silver layer performs ETL operations using Apache Spark.

b) Trip Data Processing:

- Remove duplicate trip records
- Handle missing values
- Standardize timestamp representations

c) Weather Data Processing:

- converting date format
- renaming attributes
- filtering records within 2020 to 2021

- *converting numerical attributes into appropriate data types*

d) *Holiday Data Processing: A new feature called `is_holiday` is created, where `Holiday=1` and `non-holiday=0`.*

e) *The processed trip, weather and holiday datasets will be integrated using the date attribute as the joining key, to combine daily ride count, weather information and holiday information. The integrated dataset provides a complete view of daily bike-sharing demand with external influencing factors included.*

3) *Gold Layer: Feature Engineering*

a) *In the gold layer, feature engineering will be performed to create meaningful variables for analysis.*

b) *Temporal features will be generated in this phase which include:*

- *Day of week*
- *Month*
- *Day*
- *Week of year*
- *Weekend indicator*
- *Season*

c) *Historical demand features are also created:*

- *lag_1: previous day ride count*
- *lag_7: ride count from previous week*
- *rolling_mean_7: seven-day average ride count*

d) *As a result, the analytical dataset will be generated and it will contain:*

- *ride_count*
- *weather features*
- *holiday features*
- *temporal features*
- *historical demand features*

4) *Multidimensional Database Design*

a) *To support analytical processing and visualization, the finalGold dataset is further transformed into a multidimensional database using a star schema design.*

b) *The designed star schema would be consisting of:*

- *FactBikeDemand*
- *DimDate*
- *DimWeather*
- *DimHoliday*

c) *The star schema is then connected to Power BI for interactive dashboard development to analyze bike demand patterns based on different factors.*

D. Modeling

The Modeling phase focuses on developing a predictive model to estimate daily bike-sharing demand. To enhance the analytical capability of the proposed data pipeline, a Random Forest Regression algorithm will be integrated into the system for bike-sharing demand prediction.

Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees and combines their predictions to produce more accurate and robust results [4]. The algorithm is particularly effective in handling complex

relationships between variables and can accommodate both numerical and categorical features.

The prediction model will utilize several input variables, including:

- Temperature
- Precipitation
- Rainfall
- Wind Speed
- Day of Week
- Month
- Holiday Indicator
- Historical Ride Counts

A chronological train-test split is applied with the ratio of 80% training dataset and 20% testing dataset. The trained model generates predicted bike demand values for evaluation. The generated predictions can assist bike-sharing operators in optimizing resource allocation, station management, and operational planning.

E. Evaluation

The Evaluation phase measures the performance and reliability of the developed predictive model.

Two evaluation metrics will be applied:

1) *Root Mean Square Error (RMSE)*

RMSE measures the average difference between actual and predicted ride demand values. A lower RMSE indicates that predictions are closer to actual observations.

2) *R² Score*

R² measures how well the model explains variations in bike demand. A higher R² value indicates stronger predictive capability.

F. Deployment

The Deployment phase focuses on presenting analytical outcomes through a Power BI dashboard. The multidimensional database developed from the Gold layer is connected to Power BI for interactive visualization to support analysis from temporal trends, weather impacts and holiday effects. The developed dashboard could help users to understand demand patterns and evaluate model performance.

The dashboard consists of:

- Total ride count KPI
- Average daily ride count KPI
- Maximum Demand KPI
- Actual versus predicted demand visualization
- Average demand by weekday
- Seasonal demand analysis
- Holiday and non-holiday comparison
- Weather impact analysis
- Feature importance visualization

In conclusion, the dashboard enables users to explore bike-sharing demand and utilize predictive insights for better operational planning and decision-making.

IV. SYSTEM DESIGN AND ARCHITECTURE

A. Overall System Design

The proposed system is designed to develop a bike-sharing demand prediction platform by integrating bike trip, weather and holiday datasets. In the initial phase, the system uses Apache Spark to build an ETL pipeline to process and transform all raw datasets into an analytical dataset. The analytical dataset was transferred into a star schema design for visualization purposes. After the transformation and feature engineering, the processed dataset is stored in a data lakehouse format using Parquet files.

A Random Forest Regression algorithm is implemented to predict daily bike demand based on extracted features. The bike demand and model performance results are then visualized through a Power BI dashboard to provide operators with insights into demand patterns and prediction accuracy.

B. Use Case Diagram

The use case diagram shows the interaction between system users and the proposed bike demand prediction system. The main actors are Data Engineer, who develops and manages the data pipeline, and Data User, who analyzes the demand patterns through the dashboard.

Figure 3 shows the use case diagram of the proposed bike demand prediction system.

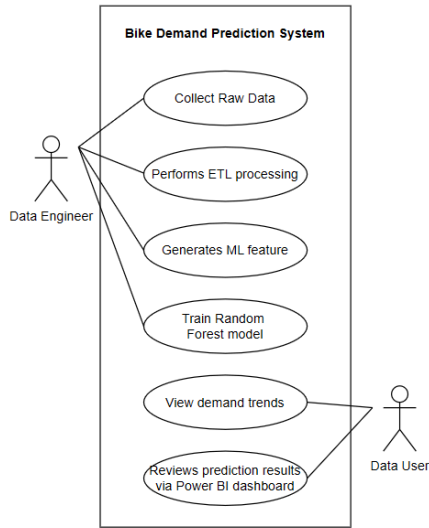


Figure 3. Use Case Diagram

C. System Architecture Diagram

The system architecture consists of five main layers, consisting of data source, ETL processing, data storage, machine learning and visualization.

The system collects data from three heterogeneous sources, Divvy bike-sharing trip records, weather data from an API and a holiday reference dataset. The data was then ingested into Bronze layers of the data lakehouse. Then, Apache Spark was used to drive the ETL processing layer, where it transforms, cleans and integrates the raw data to stores in the Silver layer.

Apache Spark was again used to engineer time-based and historical demand features to produce the analytics-ready data in the Gold layer. From Gold, the pipeline branches into two parallel outputs, where a star schema data warehouse structured for analytical querying and a Random Forest regression model was developed to generate bike demand predictions. Both outputs were transformed into Power BI to generate an interactive dashboard to end users.

Figure 4 presents the overall system architecture for the bike demand prediction system.

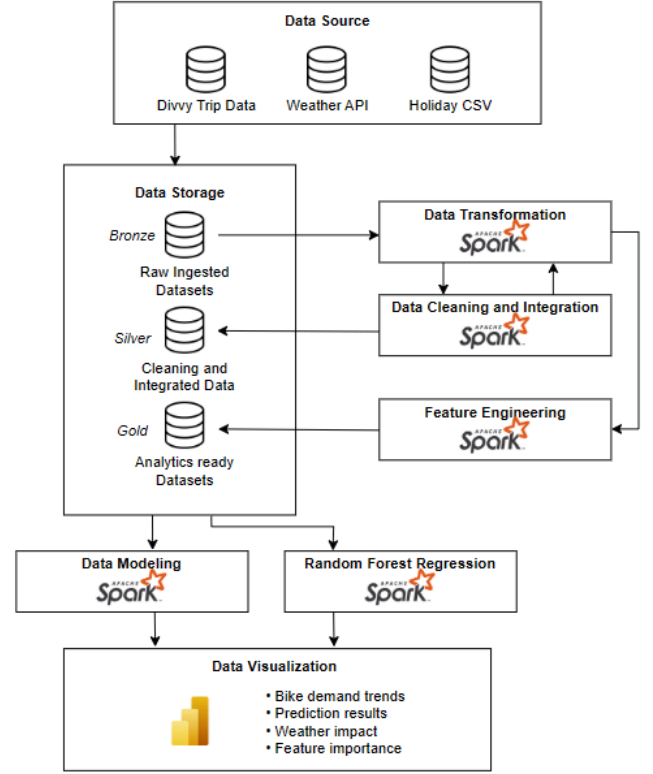


Figure 4. System Architecture Diagram

D. Database Design

The ERD represents the relationship between the original datasets used in the system. The bike trip, weather and holiday datasets are integrated using the date attribute as the common key. Figure 5 shows the ERD of the integrated data sources.

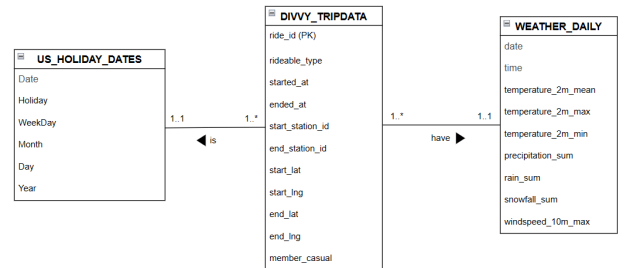


Figure 5. ERD of Data Sources

E. Multidimensional Database

A star schema is created to support analytical queries and Power BI visualisation.

The Fact_Bike_Demand table stores measurable bike demand information, including ride count and features created, such as lag_1 which represents ride count from previous day, lag_7 which represents the ride count from 7 days before and and rolling_mean_7 that calculates the average ride count of the previous 7 days. It also contains foreign keys that connect with different dimension tables.

The Dim_Date table stores time-related information such as date, month, year, day of week, weekend status and season.

The Dim_Weather table contains weather attributes such as temperature, precipitation and wind speed.

The Dim_Holiday table stores holiday information, including holiday name and holiday indicator.

Figure 6 shows the star schema design of the multidimensional database.

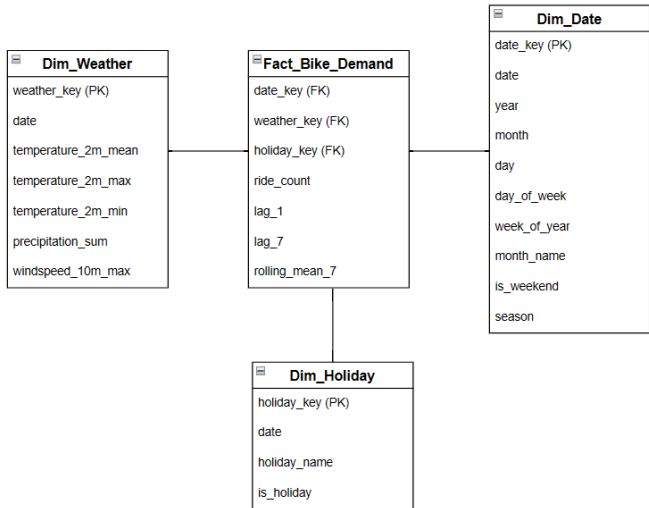


Figure 6. Star Schema Design

F. Main Interface

The main interface is developed using Power BI and two dashboard pages are created.

The first page, which is shown in Figure 7, provides bike demand analysis, specifically focusing on the external factors that influence bike demand, such as season, holiday, weather and day of week.

The second page, which is shown in Figure 8, presents Random Forest prediction results by comparing actual and predicted demand in a graph, showing model evaluation metrics and feature importance in bike demand prediction.

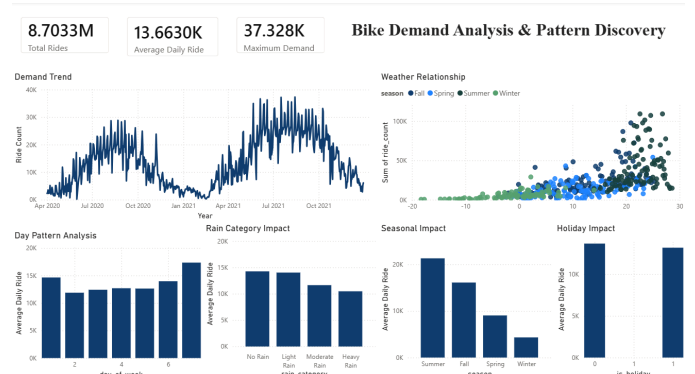


Figure 7. Bike Demand Analysis

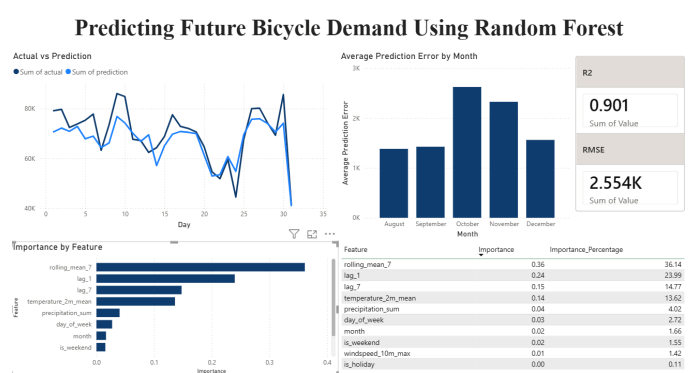


Figure 8. Random Forest Prediction Result

G. AI algorithm

The system applies Random Forest Regression to predict daily bike demand. Random Forest is selected because it can handle multiple influencing factors and capture complex relationships between all factors provided.

The input features include:

- Temperature
- Precipitation
- Wind speed
- Holiday status
- Day of week
- Weekend indicator
- Lag features
- Rolling average demand

After prediction, the model performance is evaluated using RMSE and R² score. The result is shown in Figure 8.

H. Testing Plan

System testing is carried out to ensure the correctness of each component in the proposed system. Data pipeline is tested to verify that all datasets are successfully ingested into the pipeline, invalid records were removed during data cleaning operations and all dataset joins correctly based on date attribute. Feature engineering is also validated to ensure that additional analytical attributes, including lag features, rolling average and holiday indicators, are generated for further analysis. Figure 9 shows the successful execution of the data pipeline proven with the generated analytical dataset.

date	ride_count	temperatu	temperatu	temperatu	precipitati	windspeed	holiday	is_holiday	day_of_w	month	day	week_of_y	is	week	season	lag_1	lag_7	rolling_mean_7
4/1/2020	2294	3.9	7.2	1.9	0	19.3	None	0	4	4	1	14	0	Spring	0	0	2294	
4/2/2020	2698	4.3	7.6	0.7	0	15.9	None	0	5	4	2	14	0	Spring	2294	0	2401	
4/3/2020	3532	9.6	14.8	4.5	0.1	18.6	None	0	6	4	3	14	0	Spring	2698	0	2811.333	
4/4/2020	2288	5.6	13.4	2	2.4	21.6	None	0	7	4	4	14	1	Spring	3532	0	2680.5	
4/5/2020	2887	3.2	5.5	1.3	0	19.2	None	0	1	4	5	14	1	Spring	2288	0	2721.6	
4/6/2020	2427	7.5	13.2	0.6	1.5	17.4	None	0	2	4	6	15	0	Spring	2887	0	2672.667	
4/7/2020	5185	16.8	23.9	11.6	14.3	29.2	None	0	3	4	7	15	0	Spring	2427	0	3031.571	
4/8/2020	2965	9	13.5	5.9	2.5	29.2	None	0	4	4	8	15	0	Spring	5185	2294	3041.714	
4/9/2020	1972	5.8	8.3	2	0.2	43.6	None	0	5	4	9	15	0	Spring	2965	2698	2950.857	

Figure 9. ML-ready dataset

For machine learning testing, the model is tested using unseen testing data, which consists of 20% of the analytical dataset from the Gold Layer. The trained Random Forest model achieved an RMSE of 2554.47 and R^2 score of approximately 0.90, proving that it successfully captures major patterns affecting bicycle demand. Figure 10 illustrates the comparison between the actual and predicted bicycle demand on the testing dataset, while Figure 11 shows the feature importance generated by the Random Forest model, indicating that the 7-day rolling average (rolling_mean_7), previous day's demand (lag-1) and previous week's demand (lag-7) are the most influential features for demand prediction.

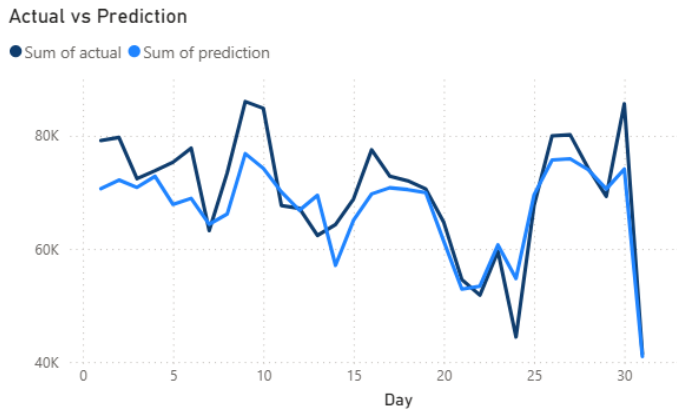


Figure 10. Actual Vs. Predicted Bicycle Demand

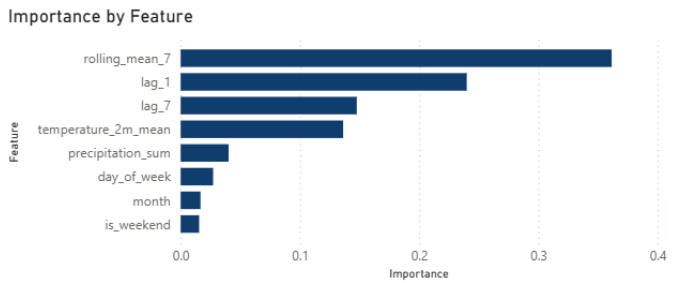


Figure 11. Random Forest Feature Importance

Finally, the Power BI dashboard is tested to ensure data visualization displays correct information so that users can analyse demand patterns effectively through charts generated.

V. CONCLUSION

This project successfully developed an Apache Spark based data engineering pipeline for bicycle demand prediction by integrating bike-sharing trip data, weather data and holiday data. The ingested data was cleaned, transformed and stored

through a Lakehouse architecture before being created into a star schema design for analytical processing and visualization in Power BI.

Feature engineering techniques, such as lag features, seasonal information and holidays indicators, were applied to improve the quality of the analytical dataset. Based on the generated features, a Random Forest Regression model was trained to predict bicycle usage demand. By achieving an R^2 score of 0.9011 and an RMSE of 2554.47, the model was proven that it can predict the demand with good accuracy. The feature importance results show that previous riding demand and weather conditions were both contributing to influential factors for demand prediction.

Finally, the Power BI dashboard visualizes historical demand patterns, weather and holiday impacts and model prediction results effectively for the data user to gain meaningful insights into bicycle usage trends. Overall, from the proposed data pipeline, an understanding on how data engineering techniques combined with machine learning can support accurate bicycle demand forecasting and provide useful information for the operators to do operational planning and resource allocation in a bike-sharing system.

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APPENDIX

Gantt Chart

